Maxime Michet

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Education

University of Toronto *Toronto, Canada*
Bachelor of Applied Science in Engineering Science – Aerospace Engineering + PEY Co-op September 2021 – Present
Minor in Mechatronics and Robotics

- **Relevant Coursework:** Mobile Robotics and Perception, Robot Modeling & Control, Spacecraft Dynamics & Control, Space Systems Design, Control Systems, Scientific Computing

Research Experience

Toronto Robotics and AI Laboratory (TRAIL) *Toronto, Canada*
Undergraduate Thesis Student September 2025 - Present

- Pursuing my undergraduate thesis on transformer-based monocular visual odometry (VO), integrating geometric priors into learning-based architectures.
- Improved performance of SOTA transformer-based VO models by incorporating pretrained depth through cross-attention.
- Currently investigating the use of transformer models as a pose refinement to classical VO pipelines.

Research Student *May 2025 - September 2025*

- Contributed to 3D anomaly detection research by improving segmentation models through 2D-3D consistency constraints and architecture upgrades.
- Developed a pipeline to augment anomaly object detection datasets by injecting anomalies into paired LiDAR-image data.
- Extended the anomaly localization in lunar orbit pipeline (ALLO) to support 3D data via depth map generation and voxel grids.

Flight Systems and Control Laboratory *Toronto, Canada*
Research Student May 2023 - September 2023

- Optimized visual-inertial localization for an autonomous drone racing (ADR) platform, achieving 4x speedup in feature matching via point/line-based tracking; validated through real-world tests.
- Built a complete autonomous drone racing platform integrating localization and flight control algorithms, now serving as the FSC lab's ADR research base.
- Contributed to the development and experimental tests of a time-optimal drone racing planner which outperformed SOTA results in planning time for optimal trajectories in drone race tracks.

Decisionics Laboratory *Toronto, Canada*
Research Student May 2022 - September 2022

- Developed novel 3D printing algorithm for FDM printers via GCode modifications, achieving 1.5x vertical strength increase over traditional methods.
- Built an in-house digital image correlation system used to measure strain and deformations of printed parts in a non-contact manner.

Industry Experience

MDA Space, Robotics and Space Operations *Brampton, Canada*
Robotics Engineering Intern May 2024 - April 2025

- Automated 20+ life cycle tests (thermal, TVAC, vibration, structural) for the Gateway External Robotics System (Canadarm3) in NASA's Artemis program.
- Simulated robotic arm configurations and created docking test plans for end-effector campaigns, verifying vision tracking of Lunar Gateway grapple fixtures.

- Designed and implemented a testing platform for data acquisition, sensor fusion, and test automation, now deployed across all test campaigns.
- Python/LabVIEW platform integrates sensor readings (thermocouples, lasers, torque cells, encoders) and controls test elements (motors, heaters).

Extra-Curricular Experience

University of Toronto Aerospace Team (UTAT) *Toronto, Canada*

Mission Operations Lead, FINCH Satellite Mission September 2021 – September 2024

- Led the mission operations team (20+ members) for FINCH, a 3U CubeSat hyperspectral mission launching in 2027.
- Developed spacecraft/ground modes, command architecture, and oversaw ground station development.
- Performed mission simulations in STK for operations planning, thermal and radiation verification.
- Initiated fault detection, isolation, and recovery efforts for system-level error analysis.
- Co-authored two papers at the Small Satellite Conference on mission design and hyperspectral denoising framework.

  Continuing as Mission Operations Advisor (September 2024 – Present)

Publications

[C.1] **Time Optimal Gate Traversing Planner for Autonomous Drone Racing**.
IEEE International Conference on Robotics and Automation (ICRA), 2024

[C.2] **Beyond the Visible: Jointly Attending to Spectral and Spatial Dimensions with HSI-Diffusion for the FINCH Spacecraft**.
Small Satellite Conference, 2024 *(group authorship)*

[C.3] **FINCH: A Blueprint for Accessible and Scientifically Valuable Remote Sensing Missions**.
Small Satellite Conference, 2022 *(group authorship)*

Honours and Awards

Best Paper on UAVs, ICRA 2024 *2024*
IEEE International Conference on Robotics and Automation

Engineering Science Research Opportunities Program - Research Award *2022*
University of Toronto

Aerospace Excellence Award *2023*
University of Toronto Aerospace Team (UTAT)

Governor General's Academic Medal *2021*
St. Marcellinus Secondary School

Skills

Programming Languages: Python, C/C++, MATLAB, JavaScript, Java, Assembly, Verilog

ML & Computer Vision: PyTorch, NumPy, pandas, OpenCV, MMsegmentation, SLURM

Software & Dev Tools: Git, Linux, Docker, CI/CD Fundamentals

Simulation & Modeling: ROS, Gazebo, Simulink, ANSYS STK, NX/TC, Fusion360

Robotics & Aerospace: Robotics, Perception, Visual Odometry, SLAM, State Estimation, Motion Planning, Spacecraft Dynamics, Space Systems Design, Orbital Mechanics, Robot Modeling & Controls

Languages: English, French